MATHS MANTHAN
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**Subject:** Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 12** **Time:** 60 Minutes
**Class:** Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM:** 30

**Document Type:** Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

**CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:**

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

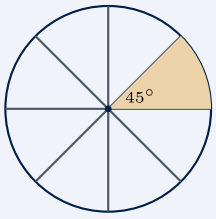
## SECTION A: MULTIPLE CHOICE

MATHS MANTHAN | (6 × 1 = 6 Marks)

**Q1.** An umbrella has 8 ribs which are equally spaced. Assuming the umbrella to be a flat circle of radius 45 cm, the angle between two consecutive ribs is: [1]

(A) 45° (B) 60° (C) 30° (D) 90°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]




**Step 1: Formula and Division of Full Angle** [0.5 Mark]
A complete circle represents a central angle of 360°. With 8 equally spaced ribs, the circle is partitioned into 8 congruent sectors.

$$\theta = \frac{360^\circ}{n} = \frac{360^\circ}{8} = 45^\circ$$

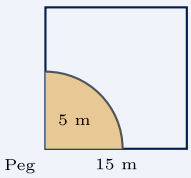
**Step 2: Conclusion** [0.5 Mark]
The central angle between two consecutive ribs is 45°.
**Correct Option:** (A) 45°

Angle between two consecutive ribs = **45°**

**Q2.** A horse is tied to a peg at one corner of a square grass field of side 15 m by means of a 5 m long rope. The area of the field the horse can graze is: (Use $\pi = 3.14$) [1]

(A) 19.625 $m^2$ (B) 78.5 $m^2$ (C) 25 $m^2$ (D) 39.25 $m^2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]




**Step 1: Sector Identification** [0.5 Mark]
At the corner vertex of a square field, the interior angle is 90°. The horse grazes a quadrant of radius $r = 5$ m. **Step 2: Area Calculation** [0.5 Mark]

$$\text{Area} = \frac{90^\circ}{360^\circ} \times \pi r^2 = \frac{1}{4} \times 3.14 \times 5^2 = \frac{1}{4} \times 3.14 \times 25 = \frac{78.5}{4} = 19.625 \text{ m}^2$$

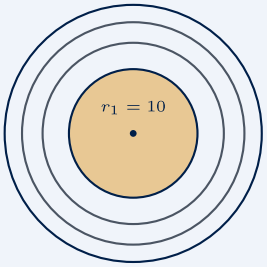
**Correct Option:** **(A) 19.625 m²**

Area horse can graze = **19.625 m²**

**Q3.** A circular disc of radius 20 cm is divided into 4 concentric parts of equal area. The radius of the innermost circle is: [1]

(A) 5 cm (B) 10 cm (C) $10\sqrt{2}$ cm (D) $5\sqrt{2}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]




**Step 1: Total Area and Partition** [0.5 Mark]
Total radius $R = 20$ cm $\implies$ Total Area $A = \pi R^2 = \pi(20)^2 = 400\pi$ cm².
Since the disc is divided into 4 concentric parts of equal area:

$$\text{Area of innermost circle } A_1 = \frac{A}{4} = \frac{400\pi}{4} = 100\pi \text{ cm}^2$$

**Step 2: Radius of Innermost Circle** [0.5 Mark]

$$\pi r_1^2 = 100\pi \implies r_1^2 = 100 \implies r_1 = 10 \text{ cm}$$

**Correct Option:** **(B) 10 cm**

Radius of innermost circle $r_1$ = **10** cm

**Q4.** If the perimeter of a square is equal to the perimeter of a circle, then the ratio of the area of the square to the area of the circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) $\pi : 4$ (B) $4 : \pi$ (C) $14 : 11$ (D) $11 : 14$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Equating Perimeters** [0.5 Mark]

Let the side of the square be $a$ and the radius of the circle be $r$.

$$\text{Perimeter of square} = \text{Circumference of circle} \implies 4a = 2\pi r \implies a = \frac{\pi r}{2}$$

**Step 2: Area Ratio Evaluation** [0.5 Mark]

$$\frac{\text{Area(square)}}{\text{Area(circle)}} = \frac{a^2}{\pi r^2} = \frac{(\pi r/2)^2}{\pi r^2} = \frac{\pi^2 r^2/4}{\pi r^2} = \frac{\pi}{4}$$

Using $\pi = \frac{22}{7}$:

$$\frac{\text{Area(square)}}{\text{Area(circle)}} = \frac{22}{7 \times 4} = \frac{22}{28} = \frac{11}{14}$$

*(Note: Both (A) $\pi : 4$ and (D) $11 : 14$ represent the same ratio; evaluated numerically using $\pi = \frac{22}{7}$, option (D) is exact.)*

**Correct Option:** (D) $11 : 14$ *(also (A) $\pi : 4$)*

$$\boxed{\frac{\text{Area of square}}{\text{Area of circle}} = \mathbf{11 : 14} \quad (\text{or } \pi : \mathbf{4})}$$

**Q5.** A pendulum swings through an angle of $30°$ and describes an arc 8.8 cm in length. The length of the pendulum is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 16.8 cm (B) 14 cm (C) 21 cm (D) 18.2 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Arc Length Formula** [0.5 Mark]

The length of the pendulum is the radius $r$ of the circular arc. Central angle $\theta = 30°$, Arc length $l = 8.8$ cm.

$$l = \frac{\theta}{360°} \times 2\pi r \implies 8.8 = \frac{30°}{360°} \times 2 \times \frac{22}{7} \times r$$

**Step 2: Solving for $r$** [0.5 Mark]

$$8.8 = \frac{1}{12} \times \frac{44}{7} \times r = \frac{11}{21} r \implies r = \frac{8.8 \times 21}{11} = 0.8 \times 21 = 16.8 \text{ cm}$$

**Correct Option:** (A) 16.8 cm

$$\boxed{\text{Length of the pendulum} = \mathbf{16.8} \text{ cm}}$$

**Q6.** **Assertion (A):** If a wire of given length is bent first into a square and then into a circle, the area enclosed by the circle is greater than the area enclosed by the square.
**Reason (R):** For a fixed perimeter, the ratio of the area of a square to that of a circle is $\pi : 4 \approx 11 : 14 < 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Verification of Assertion (A)** [0.5 Mark]

For a fixed wire length $L$, perimeter $= L$.

$$\text{Area(square)} = \left(\frac{L}{4}\right)^2 = \frac{L^2}{16}, \qquad \text{Area(circle)} = \pi\left(\frac{L}{2\pi}\right)^2 = \frac{L^2}{4\pi}$$

Since $4\pi \approx 12.57 < 16$, $\frac{L^2}{4\pi} > \frac{L^2}{16}$. Hence, the circle encloses more area. Assertion (A) is **True**.

**Step 2: Verification of Reason (R)** [0.5 Mark]

$$\frac{\text{Area(square)}}{\text{Area(circle)}} = \frac{4\pi}{16} = \frac{\pi}{4} \approx \frac{11}{14} < 1$$

This proves that circle area is greater than square area. Thus, (R) is **True** and is the correct explanation of (A).

**Correct Option:** **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

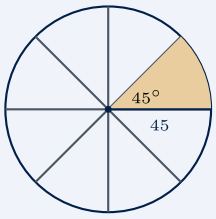
**Option (A) is correct**

## SECTION B: Very Short Answer Type

MATHS MANTHAN | **(2 × 2 = 4 Marks)**

**Q7.** An umbrella has 8 ribs which are equally spaced. Assuming the umbrella to be a flat circle of radius 45 cm, find the area between two consecutive ribs. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

► **STEP-BY-STEP SOLUTION:** [2 Marks]




**Step 1: Formula and Central Angle** [1 Mark]

The flat circle has radius $r = 45$ cm. With 8 equally spaced ribs, the circle is partitioned into 8 congruent sectors.

$$\theta = \frac{360^\circ}{8} = 45^\circ$$

$$\text{Area of one sector} = \frac{1}{8}\pi r^2$$

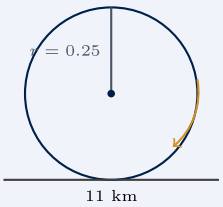
**Step 2: Substitution and Calculation** [1 Mark]

$$\text{Area} = \frac{1}{8} \times \frac{22}{7} \times 45 \times 45 = \frac{11 \times 2025}{28} = \frac{22275}{28} \approx 795.54 \text{ cm}^2$$

$$\boxed{\text{Area between two consecutive ribs} = \frac{22275}{28} \text{ cm}^2 \approx \mathbf{795.54} \text{ cm}^2}$$

**Q8.** The radius of a wheel is 0.25 m. Find the number of complete revolutions it will make to travel a distance of 11 km. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]




**Step 1: Distance Covered in One Revolution** [1 Mark]

Radius $r = 0.25$ m $= \frac{1}{4}$ m.

$$\text{Circumference } C = 2\pi r = 2 \times \frac{22}{7} \times \frac{1}{4} = \frac{11}{7} \text{ m}$$

**Step 2: Total Revolutions Calculation** [1 Mark]

Total distance to cover $D = 11$ km $= 11 \times 1000 = 11000$ m.

$$\text{Number of revolutions } n = \frac{D}{C} = \frac{11000}{11/7} = 11000 \times \frac{7}{11} = 1000 \times 7 = 7000$$

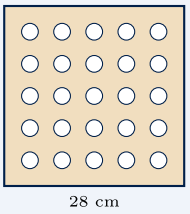
Number of complete revolutions = **7000**

## SECTION C: Short Answer Type

MATHS MANTHAN | **(2 × 3 = 6 Marks)**

**Q9.** A square metal drain plate has side 28 cm. Inside the plate, 25 identical circular holes of diameter 2.8 cm each are drilled to drain water. Find the remaining area of the plate. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]




**Step 1: Area of Square Drain Plate** [1 Mark]

Side of square plate $a = 28$ cm.

$$\text{Area of square plate} = a^2 = 28^2 = 784 \text{ cm}^2$$

**Step 2: Area of 25 Circular Holes** [1 Mark]

Diameter of each hole $d = 2.8$ cm $\implies$ radius $r = 1.4$ cm.

$$\text{Area of 1 hole} = \pi r^2 = \frac{22}{7} \times 1.4 \times 1.4 = \frac{22}{7} \times 1.96 = 22 \times 0.28 = 6.16 \text{ cm}^2$$

$$\text{Total area of 25 holes} = 25 \times 6.16 = 154 \text{ cm}^2$$

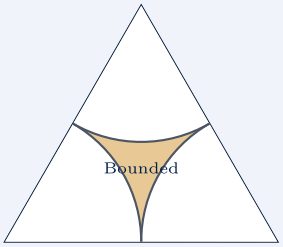
**Step 3: Remaining Area of the Plate** [1 Mark]

$$\text{Remaining Area} = \text{Area of square} - \text{Total area of holes} = 784 - 154 = 630 \text{ cm}^2$$

Remaining area of the plate = **630 cm²**

**Q10.** An equilateral triangle has side 24 cm. From each vertex, an arc of radius 12 cm is drawn touching the other two arcs. Find the area of the bounded space between the arcs. (Use $\pi = 3.14$, $\sqrt{3} = 1.732$) [3]

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]




**Step 1: Area of Equilateral Triangle** [1 Mark]

Side of equilateral triangle $a = 24$ cm.

$$\text{Area}(\triangle ABC) = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 24^2 = \frac{\sqrt{3}}{4} \times 576 = 144\sqrt{3}$$

$$= 144 \times 1.732 = 249.408 \text{ cm}^2$$

**Step 2: Area of Three Sectors at the Vertices** [1 Mark]

Since arcs touch each other pairwise, $r = \frac{24}{2} = 12$ cm. Each interior angle $\theta = 60°$.

$$\text{Total Area of 3 sectors} = 3 \times \left(\frac{60°}{360°} \times \pi r^2\right) = 3 \times \frac{1}{6} \times \pi r^2 = \frac{1}{2}\pi r^2$$

$$= \frac{1}{2} \times 3.14 \times 12^2 = \frac{1}{2} \times 3.14 \times 144 = 3.14 \times 72 = 226.08 \text{ cm}^2$$

**Step 3: Area of Bounded Space** [1 Mark]

$$\text{Area} = \text{Area}(\triangle ABC) - \text{Area of 3 sectors} = 249.408 - 226.08 = 23.328 \text{ cm}^2$$

Area of bounded space = **23.328 cm² ≈ 23.33 cm²**

## SECTION D: Long Answer / HOTS Type — MATHS MANTHAN | (2 × 5 = 10 Marks)

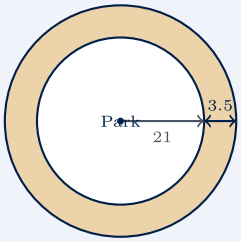
**Q11.** A circular municipal park of radius 21 m is surrounded by an outer jogging path 3.5 m wide. A protective steel fence is erected along the outer boundary of the path, and the entire path is graveled. [5]

(a) Find the inner and outer radii and circumferences of the path.

(b) Calculate the surface area of the jogging path.

(c) If fencing costs Rs. 25 per metre and graveling costs Rs. 16 per m², calculate the total cost of the improvement project. $\left(\text{Use } \pi = \frac{22}{7}\right)$

► **STEP-BY-STEP SOLUTION:** [5 Marks]




**(a) Radii and Circumferences** [1.5 Marks]

Inner radius $r = 21$ m, Width of path $w = 3.5$ m.

$$\text{Outer radius } R = r + w = 21 + 3.5 = 24.5 \text{ m} = \frac{49}{2} \text{ m}$$

$$\text{Inner Circumference } C_{\text{inner}} = 2\pi r = 2 \times \frac{22}{7} \times 21 = 44 \times 3 = 132 \text{ m}$$

$$\text{Outer Circumference } C_{\text{outer}} = 2\pi R = 2 \times \frac{22}{7} \times \frac{49}{2} = 22 \times 7 = 154 \text{ m}$$

**(b) Surface Area of Jogging Path** [1.5 Marks]

$$\text{Area of path} = \pi(R^2 - r^2) = \frac{22}{7}(24.5^2 - 21^2) = \frac{22}{7} \times (24.5 - 21)(24.5 + 21)$$

$$= \frac{22}{7} \times 3.5 \times 45.5 = 22 \times 0.5 \times 45.5 = 11 \times 45.5 = 500.5 \text{ m}^2$$

**(c) Total Project Cost** [2 Marks]

- Fencing along outer boundary: $\text{Cost}_{\text{fence}} = 154 \text{ m} \times \text{Rs. } 25 = \text{Rs. } 3850$
- Graveling path surface: $\text{Cost}_{\text{gravel}} = 500.5 \text{ m}^2 \times \text{Rs. } 16 = \text{Rs. } 8008$
- Total cost $= 3850 + 8008 = \text{Rs. } 11858$

(a) $R = 24.5$ m, $C_{\text{in}} = 132$ m, $C_{\text{out}} = 154$ m; (b) Area $= 500.5$ m$^2$; (c) Total Cost = Rs. **11858**

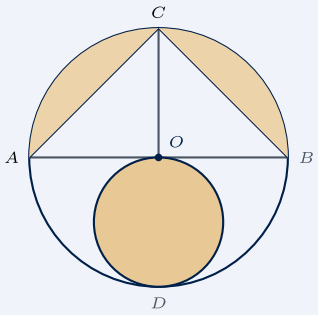
**Q12.** In a circle of radius 28 cm, two diameters $AB$ and $CD$ are perpendicular to each other. With $OD$ as diameter, a smaller circle is drawn. [5]

(a) Find the radius and area of the smaller circle.

(b) Find the area of $\triangle ABC$ formed by diameter $AB$ and vertex $C$.

(c) Calculate the area of the shaded region consisting of the small circle and the two segments of the larger circle bounded by $AC, BC$ and arc $ACB$. $\left(\text{Use } \pi = \frac{22}{7}\right)$

► **STEP-BY-STEP SOLUTION:** [5 Marks]




**(a) Radius and Area of Smaller Circle** [1.5 Marks]

Radius of large circle $R = 28$ cm. Diameter of smaller circle $= OD = R = 28$ cm.

$$\text{Radius of small circle } r = \frac{OD}{2} = \frac{28}{2} = 14 \text{ cm}$$

$$\text{Area of small circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 22 \times 28 = 616 \text{ cm}^2$$

**(b) Area of $\triangle ABC$** [1.5 Marks]

Base $AB = 2R = 2 \times 28 = 56$ cm, Altitude $OC = R = 28$ cm.

$$\text{Area}(\triangle ABC) = \frac{1}{2} \times AB \times OC = \frac{1}{2} \times 56 \times 28 = 28 \times 28 = 784 \text{ cm}^2$$

**(c) Area of Total Shaded Region** [2 Marks]

$$\text{Area of semicircle } ACB = \frac{1}{2}\pi R^2 = \frac{1}{2} \times \frac{22}{7} \times 28^2 = 11 \times 112 = 1232 \text{ cm}^2$$

Area of 2 segments bounded by $AC, BC$ and arc $ACB = 1232 - 784 = 448$ cm$^2$

Total Shaded Area = Area(small circle)+Area(2 segments) $= 616 + 448 = 1064$ cm$^2$

(a) Small Circle Area $= 616$ cm$^2$; (b) Area$(\triangle ABC) = 784$ cm$^2$; (c) Total Shaded Area $=$ **1064 cm$^2$**

## SECTION E: Case-Based / Applied Situation MATHS MANTHAN | (1 × 4 = 4 Marks)

**Q13. Automotive Speedometer and Wheel Transmission Physics**

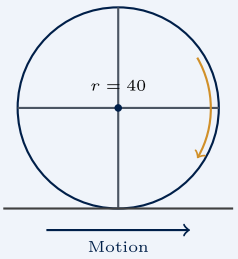
A vehicle instrumentation engineer tests the odometer and speedometer accuracy of a car. The car wheels have an outer diameter of $d = 80$ cm. The vehicle is driven at a steady cruising speed of 66 km/h for a test duration of 10 minutes.

(i) Calculate the distance traveled by the car in one complete revolution of a wheel. [1]

(ii) Find the total distance traveled by the car in 10 minutes. [1]

(iii) Calculate the total number of complete revolutions each wheel makes in 10 minutes, **OR** find the number of revolutions per minute (RPM) of the wheels at this cruising speed. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

► **STEP-BY-STEP SOLUTION:** [4 Marks]




**(i) Distance Covered in One Complete Revolution** [1 Mark]

Diameter $d = 80$ cm $\implies$ radius $r = 40$ cm.

$$\text{Distance in 1 revolution} = \text{Circumference } C = \pi d = \frac{22}{7} \times 80 = \frac{1760}{7} \text{ cm} \approx 251.43 \text{ cm} = 2.5143 \text{ m}$$

**(ii) Total Distance Traveled in 10 Minutes** [1 Mark]

$$\text{Speed } v = 66 \text{ km/h}, \qquad \text{Time } t = 10 \text{ minutes} = \frac{10}{60} \text{ hour} = \frac{1}{6} \text{ hour}$$

$$\text{Total Distance } D = v \times t = 66 \times \frac{1}{6} = 11 \text{ km} = 11000 \text{ m} = 1100000 \text{ cm}$$

**(iii) Total Revolutions in 10 Minutes (OR Wheel RPM)** [2 Marks]

$$\text{Total revolutions } N = \frac{\text{Total Distance}}{\text{Circumference}} = \frac{1100000}{1760/7} = \frac{1100000 \times 7}{1760} = \frac{7700000}{1760} = 4375$$

**OR (Revolutions Per Minute - RPM):**

$$\text{RPM} = \frac{\text{Total revolutions}}{\text{Time in minutes}} = \frac{4375}{10} = 437.5 \text{ RPM}$$

*(Alternatively: Speed* $= \frac{1100000 \text{ cm}}{10 \text{ min}} = 110000$ *cm/min* $\implies$ *RPM* $= \frac{110000}{1760/7} = 437.5$*.)*

**(i)** $\frac{1760}{7}$ cm $\approx$ **2.51** m; **(ii) 11** km; **(iii) Total Revolutions = 4375** (**OR RPM = 437.5**)

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